STRAVA FITNESS DATA ANALYTICS CASE STUDY

By

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For

Labmentix Intership 3rd Project

1. Executive Summary

This report represents as comprehensive analysis of Fitbit user behavior data across 33 tracked users over a one-month period. The analysis examines physical activity patterns, calorie expenditure, sleep quality, heart rate zones, and feature engagement to surface actionable product and health insights.

Key findings reveal that while most users are highly engaged with th platform (82% tracking 28-31 days), the majority fall short of key health benchmarks, average daily steps are 17% below the recommended 10,000 goal, average sleep duration is nearly one hour below the recommended 8 hours, and sedentary time averages an alarming 15.9 hours per day. These gaps represent significant opportunities for product intervention.

2. MySQL Workflow

2.1 Database and Schema Creation

A dedicated database named strava_app was created to house all project tables. Fifteen tables were defined with explicit column names and data types to match the structure of 18 source CSV files. Three files minuteCaloriesNarrow, minuteIntensitiesNarrow and minuteMETsNarrow could not be fully imported due to system memory constraints and were excluded from analysis. Their data was adequately covered by the wide-format equivalents and hourly tables already loaded.

A critical schema decision was made during this phase: all Id columns were defined as BIGINT rather than the default INT. Fitbit user IDs are 10-digit numbers (e.g. 8877689391) that exceed the maximum value of the standard MySQL INT data type (2,147,483,647). Using INT caused import failures with Error Code 1264: Out of range value for column 'Id', which was resolved by switching all ID columns to BIGINT.

2.2 Data Integrity Check

A comprehensive set of integrity checks was run across all tables before any analysis to ensure data quality and identify potential issues.

- Row Count Verification
  All 15 tables were queried for total row counts and verified against expected values from the source files.
- Unique User Counts
  COUNT(DISTINCT Id) was run across all tables to identify participation levels. Key findings:
  - Daily activity, hourly, and minute tables: 33 users
  - Sleep tracking: 24 users
  - Weight logging: 8 users
  - Heart rate: 14 users

- **NULL Value Checks**
  All columns in daily_activity and weight_log were checked for NULL values. The Fat column in weight_log was confirmed to have 65 of 67 values as NULL that made it practically unusable for analysis. All other core activity columns returned zero NULLs.

- **Duplicate Record Detection**
  Duplicate checks were run on (Id, Date) combinations across key tables. Three duplicate records were found in sleep_day — the same user logged sleep twice on the same night. No duplicates were found in daily_activity.

- **Out-of-Range Value Checks**
  Minimum and maximum values were verified for TotalSteps, Calories, SedentaryMinutes, VeryActiveMinutes and TotalDistance. All values fell within physiologically plausible ranges.

- **Zero Step Days**
  77 days (8.2% of all records) had zero total steps across 15 users — indicating days the tracker was not worn or not moved. These were flagged for exclusion from activity analysis.

- **Full Sedentary Days**
  79 records showed SedentaryMinutes >= 1440 which was the equivalent of 24 hours of inactivity observed across 17 users. These records were identified as likely tracker-off days where the device was worn but not moved.

- **Sleep State Validation**
  The minute_sleep table uses a numeric value code (1 = Asleep, 2 = Restless, 3 = Awake) to record minute-level sleep states. A distinct value check confirmed only values 1, 2 and 3 were present with no invalid entries.

2.3 Data Cleaning

- **Sleep Duplicate Removal**: The three duplicate sleep records identified were removed by creating a cleaned table sleep_day_clean retaining only the first occurrence of each (Id, SleepDay) pair. The row count reduced from 413 to 410 after duplicate removal.

- **Date Format Standardisation**: All date and datetime columns in the raw CSV files were stored as strings in the format M/D/YYYY and M/D/YYYY H:MM:SS AM/PM. These were converted to standardised MySQL DATE and DATETIME types using **STR_TO_DATE()** and added as new clean columns. This was repeated on columns daily_activity, hourly_steps, hourly_calories, hourly_intensities, sleep_day_clean, weight_log and heartrate_seconds.

- **Weekday Column Addition:** A DayOfWeek column was derived from each cleaned date column using **DAYNAME()** to enable weekday-level grouping and analysis.

- **Hour of Day Column Addition:** A HourOfDay column was derived from all hourly and heartrate datetime columns using **HOUR()** to enable hour-level aggregation.

- **Total Active Minutes Derivation:** A TotalActiveMinutes column was added to daily_activity by summing the three active intensity categories.

- **Sleep Duration and Onset Derivation:** Two derived columns were added to sleep_day_clean:

  - SleepHours = TotalMinutesAsleep / 60.0 — for human-readable sleep duration

  - MinsToFallAsleep = TotalTimeInBed - TotalMinutesAsleep — to measure sleep onset time

3. Python Workflow

The Python component of this project was used to perform **data analysis and visualization** on the processed datasets. The workflow focuses on loading prepared datasets, analyzing user activity metrics, and generating visual insights to support the project findings.

- **Importing Required Libraries**

  The workflow begins by importing the necessary Python libraries used for data manipulation and visualization.

  - Pandas – for data loading and data manipulation.
  - Matplotlib – for creating charts and plots.
  - Seaborn – for advanced statistical visualizations.

  These libraries enable efficient analysis and graphical representation of the fitness tracking data.

- **Data Loading**
  The processed datasets generated from earlier SQL transformations were imported into Python using Pandas.
  The following CSV files were loaded for analysis:
  - feature_adoption_summary.csv
  - activity_level_classification.csv
  - per_user_BMI_vs_sleep.csv
  - heartrate_zone_per_user.csv
  - user_segmentation.csv

  Each dataset represents a specific analytical perspective of the user activity data, such as feature usage, activity levels, sleep patterns, heart rate zones, and user segmentation.

- **Feature Adoption Analysis**
  The feature adoption dataset was analyzed to calculate the percentage of users who use different tracking features such as:
  - Sleep tracking
  - Weight tracking
  - Other health metrics

  This was done by calculating the proportion of users using each feature relative to the total number of users. The results help understand which features are most widely used by users.

- **Activity Level Analysis**
  User activity levels were analyzed using the activity_level_classification dataset.
  A violin plot was created to visualize the distribution of average daily calories burned across different activity levels. This allows comparison of calorie expenditure between groups such as:
  - Sedentary users
  - Moderately active users
  - Highly active users

  The visualization highlights how energy expenditure varies with user activity levels.

- **Sleep vs Sedentary Behaviour Analysis**
  To examine lifestyle behavior patterns, the dataset per_user_BMI_vs_sleep.csv was analyzed. A joint regression plot was generated to study the relationship between:
  - Average sedentary minutes per day
  - Average sleep hours per day

  This analysis helps identify whether increased sedentary time correlates with changes in sleep duration.

- **Heart Rate Zone Distribution**
  The dataset heartrate_zone_per_user.csv was used to analyze heart rate intensity zones for users. The analysis calculated the percentage of time users spend in different heart rate zones:
  - Zone 1 – Rest
  - Zone 2 – Fat burn
  - Zone 3 – Aerobic
  - Zone 4 – Cardio
  - Zone 5 – Peak

  A stacked visualization was used to compare these zones across users and understand exercise intensity patterns.

- **User Segmentation Analysis**
  The user_segmentation.csv dataset was used to evaluate how different user activity groups perform relative to the 10,000 daily steps goal**.** A pivot table was created to calculate the average percentage of days users met the 10K step goal for each activity level.
  A heatmap visualization was generated to clearly display the relationship between activity level and goal achievement.

4. Overview (Key Metrics)

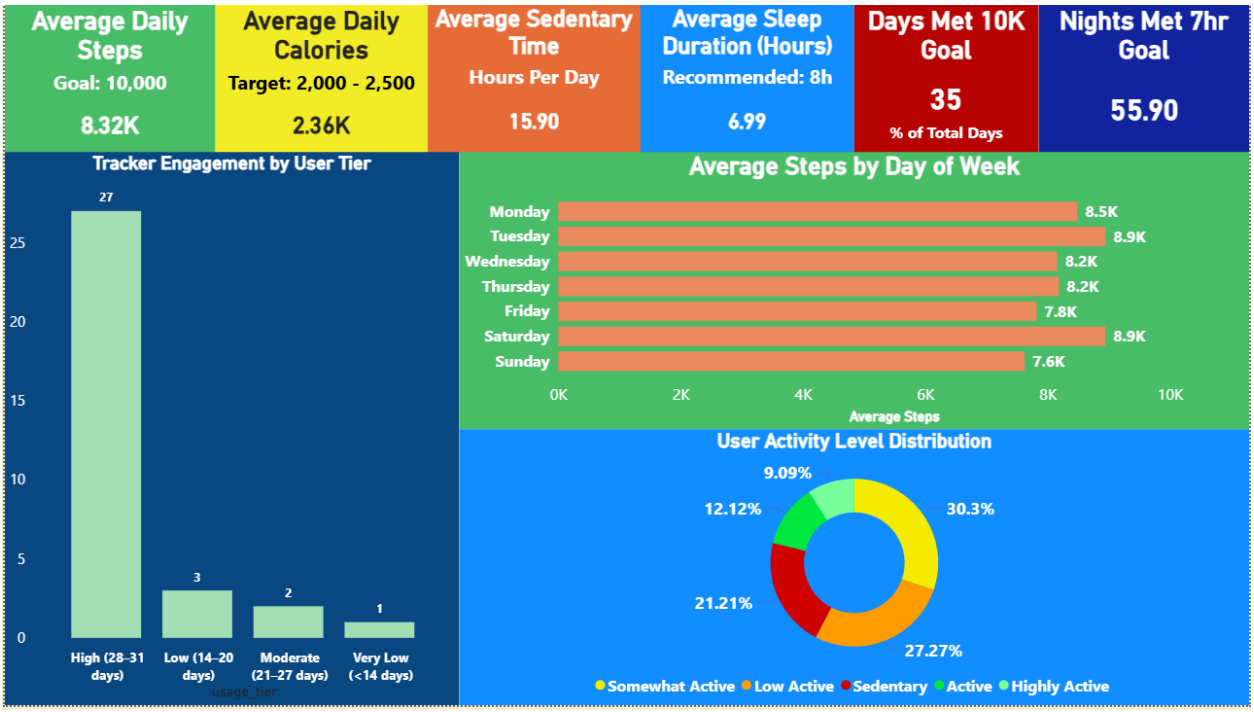


Figure 1: Main KPI Dashboard: Steps, Calories, Sedentary Time, Sleep & Goal Achievement

The overview dashboard confirms that while calorie burn targets are being met (2,360 average vs 2,000 to 2,500 target range), step counts and sleep are consistently below the recommended target. Tracker engagement is good with 27 of 33 users in high usage tier (28-31 days).

Activity level distribution is relatively balanced across the 5 segments, with sedentary and somewhat active users each representing approximately 27=30% of the user base. This indicates that significant room for behavioural nudging.

5. Physical Activity Analysis

5.1 Steps and Calories by Day of Week

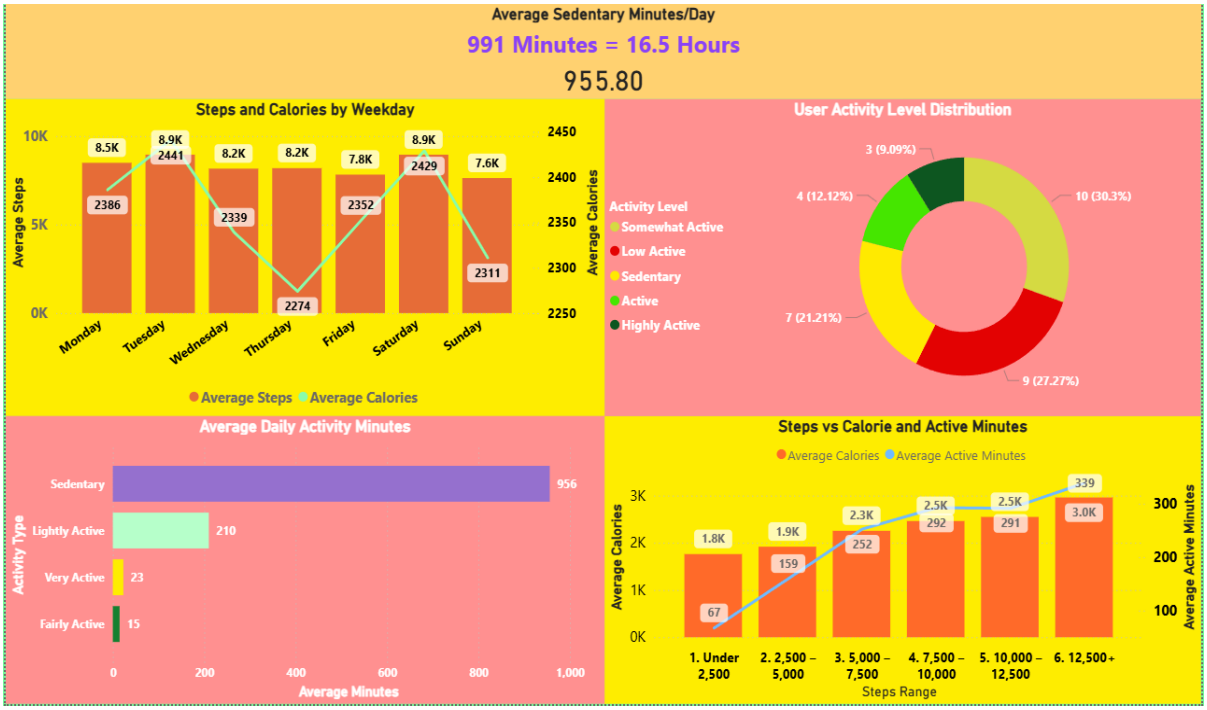
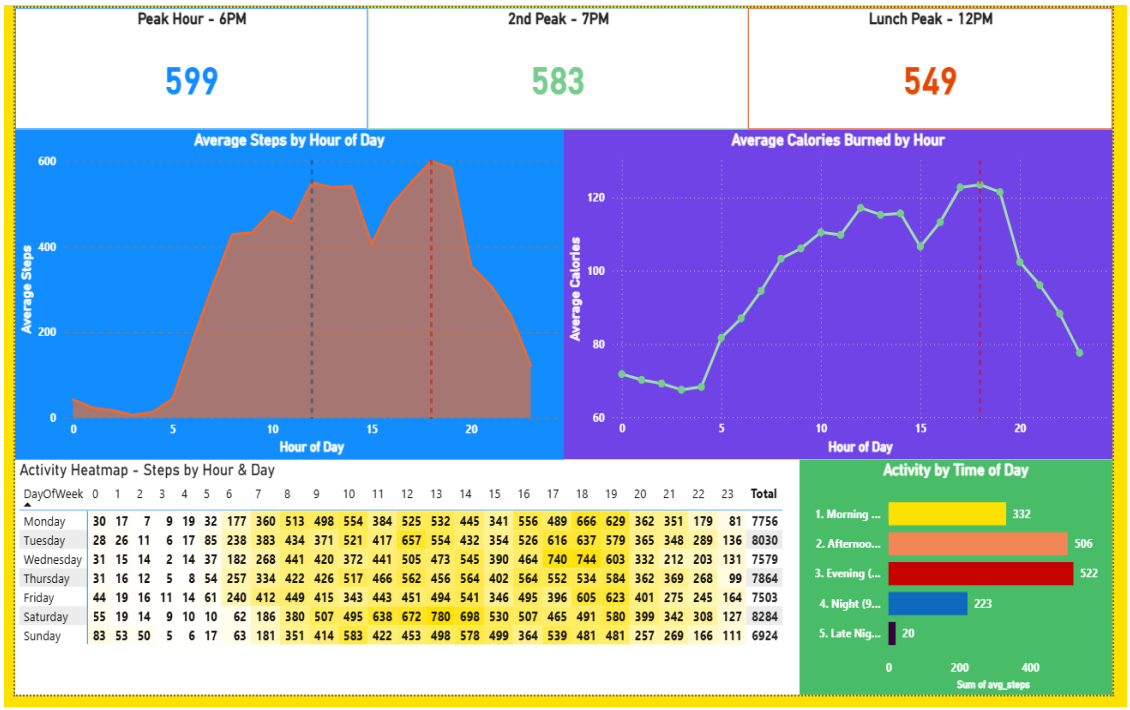


Figure 2: Activity & Engagement Dashboard: Steps/Calories by Weekday, Activity Distribution

Step counts remain relatively consistent across the week, ranging from 7.6K (Sunday) to 8.9K (Tuesday and Saturday). Notably, Saturday rivals weekdays for step activity, this suggests that users are active during leisure time and not just on work commutes.

Thursday shows the lowest step count (8.2K) and the lowest average calories (2,274), indicating a mid-week energy dip. Calorie burn correlates closely with steps across all days, validating step count as a primary activity proxy.

5.2 Hourly Activity Patterns



DayOfWeek	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	Total
Monday	30	17	7	9	19	32	177	360	513	498	554	384	525	532	445	341	556	489	666	629	362	351	179	81	7756
Tuesday	28	26	11	6	17	85	238	383	434	371	521	417	657	554	432	354	526	616	637	579	365	348	289	136	8030
Wednesday	31	15	14	2	14	37	182	268	441	420	372	441	505	473	545	390	464	740	744	603	332	212	203	131	7579
Thursday	31	16	12	5	8	54	257	334	422	426	517	466	562	456	564	402	564	552	534	584	362	369	268	99	7864
Friday	44	19	16	11	14	61	240	412	449	415	343	443	451	494	541	346	495	396	605	623	401	275	245	164	7503
Saturday	55	19	14	9	10	10	62	186	380	507	495	638	672	780	698	530	507	465	491	580	399	342	308	127	8284
Sunday	83	53	50	5	6	17	63	181	351	414	583	422	453	498	578	499	364	539	481	481	257	269	166	111	6924

Figure 3: Hourly Activity Dashboard: Peak Steps by Hour, Calorie Burn, Activity Heatmap

The hourly analysis reveals a tri modal activity pattern with three distinct peaks:

- Morning peak at 12PM (549 average steps), that is likely lunch walk activity
- Afternoon peak at 6PM (599 average steps), this is the single highest activity hour
- Secondary evening peak at 7PM (583 average steps), which indicates post work exercise

Early morning hours (0-6AM) are consistently almost zero, with activity increasing from around 7AM. Evening activity (6-7PM) dominates, representing the optimal window for push notification campaigns or challenge activations.

The activity heatmap by day and hour confirms this pattern hold across all weekdays, with Saturday showing elevated midday activity compared to weekdays.

5.3 Calorie Distribution by Activity Segment

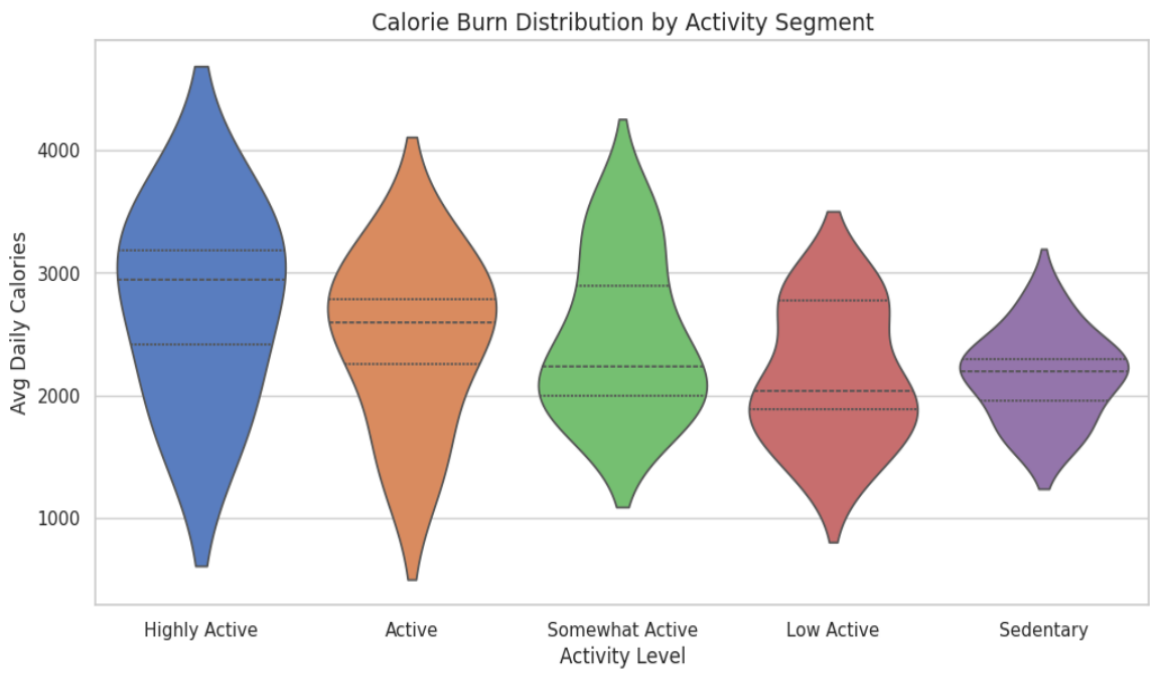


Figure 4: Calorie Burn Distribution by Activity Segment (Violin Plot)

The average daily calories by activity level reveal important distributional differences. Highly active users show the widest spread (roughly 750-4,600 calories), reflecting high intragroup variability. This suggests that the 'Highly Active' label captures a diverse population including both endurance athletes and highly mobile professionals.

Sedentary and Low Active users show tighter, lower range distributions concentrated around 2,000 – 2,500 calories. The somewhat Active group shows a bimodal shape. This suggests two distinct sub populations within this segment which is a potential signal for further segmentation.

5.4 Step Goal Consistency

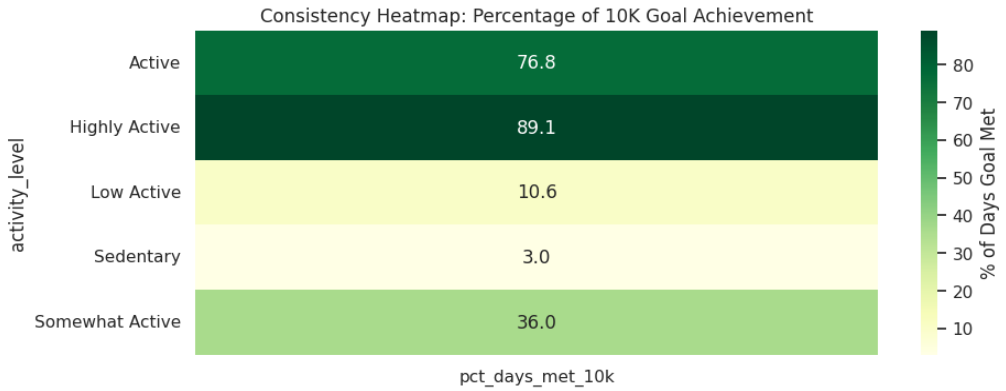


Figure 5: 10K Goal Achievement Heatmap by Activity Level

10K step goal achievement is strongly stratified by activity level. Highly active users achieve the goal on 89.1% if tracked days and Active users on 76.8%.

In contrast, Sedentary users meet the goal only 3% of the days and Low Active users only 10.6%. Somewhat Active users achieve it 36% of the time.

These patterns support a tiered notification strategy, rather than a one size fits all daily reminder, messaging should be calibrated to each user's achievable threshold to drive incremental improvement.

6. Sleep Quality Analysis

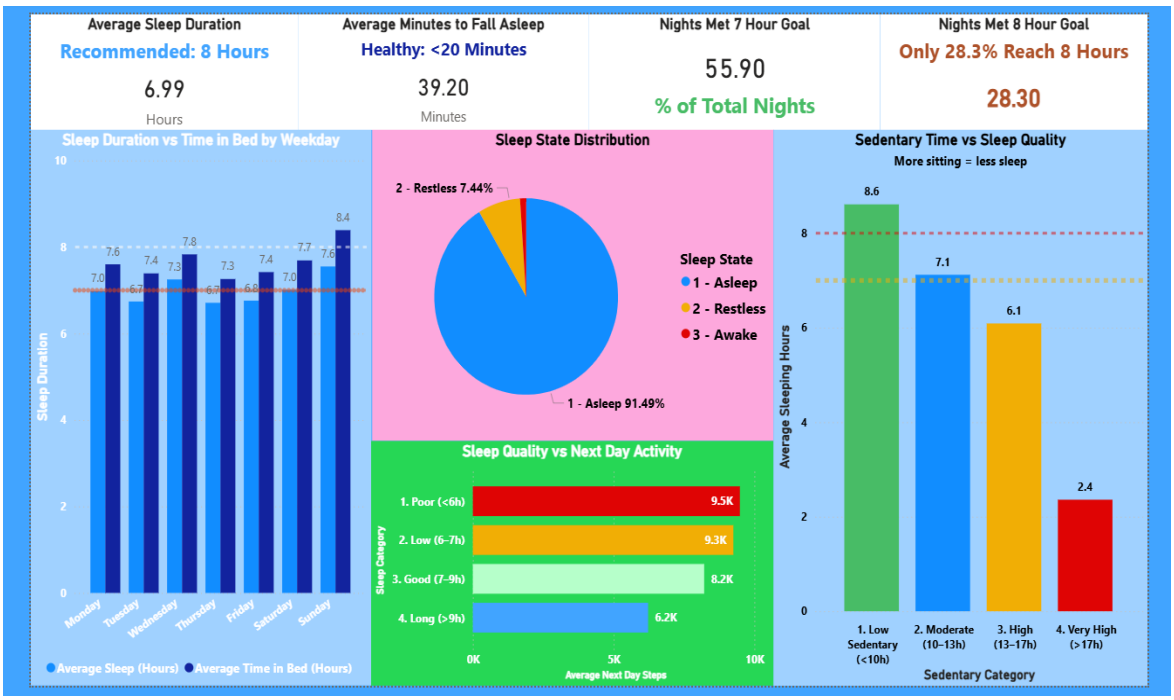


Figure 6: Sleep Dashboard: Duration by Weekday, Sleep States, Sleep vs. Next-Day Activity

Sleep is a major area of concern across the user base. At an average of 6.99 hours, users fall approximately one hour short of the recommended 8 hours. Only 28.3% of users meet the 8-hour goal on any given night, while 55.9% meet the more modest 7-hour threshold.

6.1 Sleep duration by Day of Week

Sunday registers the highest average sleep duration (7.6 hours time in bed, 8.4 hours including pre-sleep), while weekdays cluster between 6.7-7.4 hours. The gap between time in bed and actual sleep duration is consistent at approximately 0.5-0.7 hours, implying users spend meaningful time awake in bed. This aligns with the 39.2 minutes average time to fall asleep metric.

6.2 Sleep State Distribution

91.49% of tracked sleep time is classified as ‘Asleep’ with 7.44% Restless and only 1.07% Awake. While the headline figure appears healthy, the restlessness rate and extended sleep onset time suggest sleep quality issues that the raw duration metric does not fully capture.

6.3 Sleep Quality vs Next Day Performance

A compelling productivity signal emerges from the sleep-activity relationship. Users with poor sleep (under 6 hours) actually record the highest next-day step counts (9,500 average), followed by Low sleep (6-7 hours) at 9.300 steps. Good sleepers (7-9 hours) average 8,200 steps, and Long sleepers (9+ hours) average only 6,200 steps.

The counterintuitive pattern likely reflects lifestyle confounders: Active individuals may sleep less due to demanding schedules yet remain highly mobile. Long sleep may indicate health issues or sedentary recovery. This warrants deeper investigation before drawing causal conclusions.

6.4 Sedentary Time and Sleep Quality

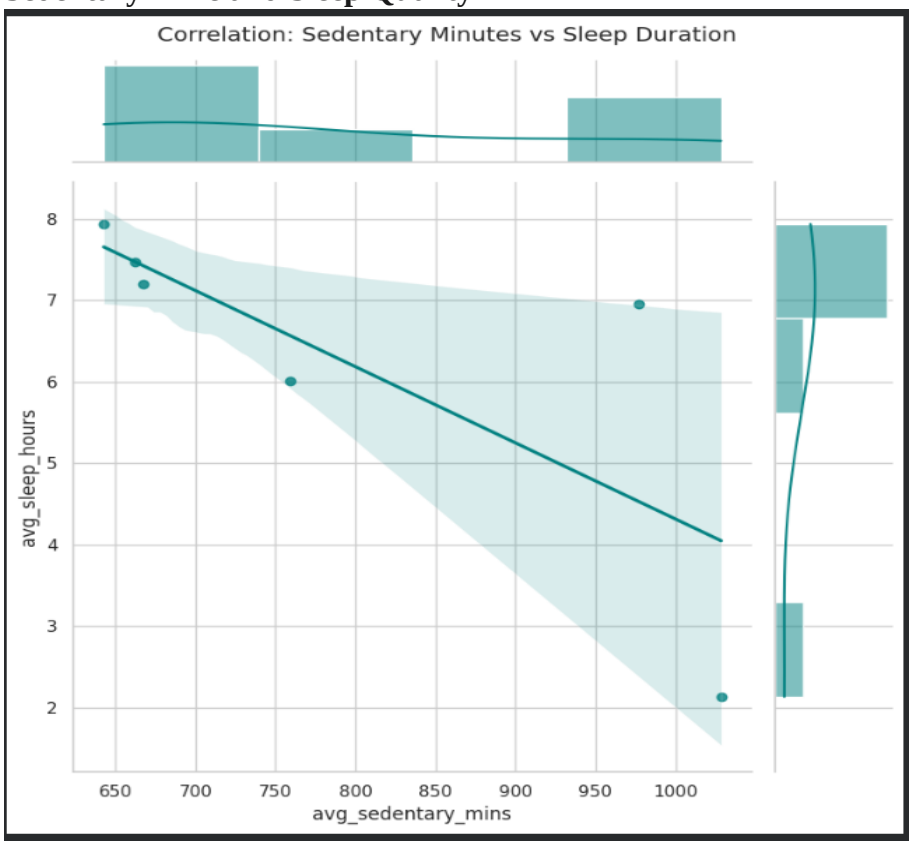


Figure 7: Correlation: Sedentary Minutes vs. Average Sleep Duration

The scatter plot reveals a notable negative correlation between daily sedentary minutes and sleep duration. Users with low sedentary time (under 10 hours) average 8.6 hours of sleep, while those with very high sedentary time (17+ hours) average only 2.4 hours.

The regression line confirms this trend, though the small sample size (n=33) means the results should be treated as directional.The finding supports that sedentary behaviour as a potential level for sleep improvement is a key insight for in-app coaching features.

7. Heart Rate and Physiological Analysis

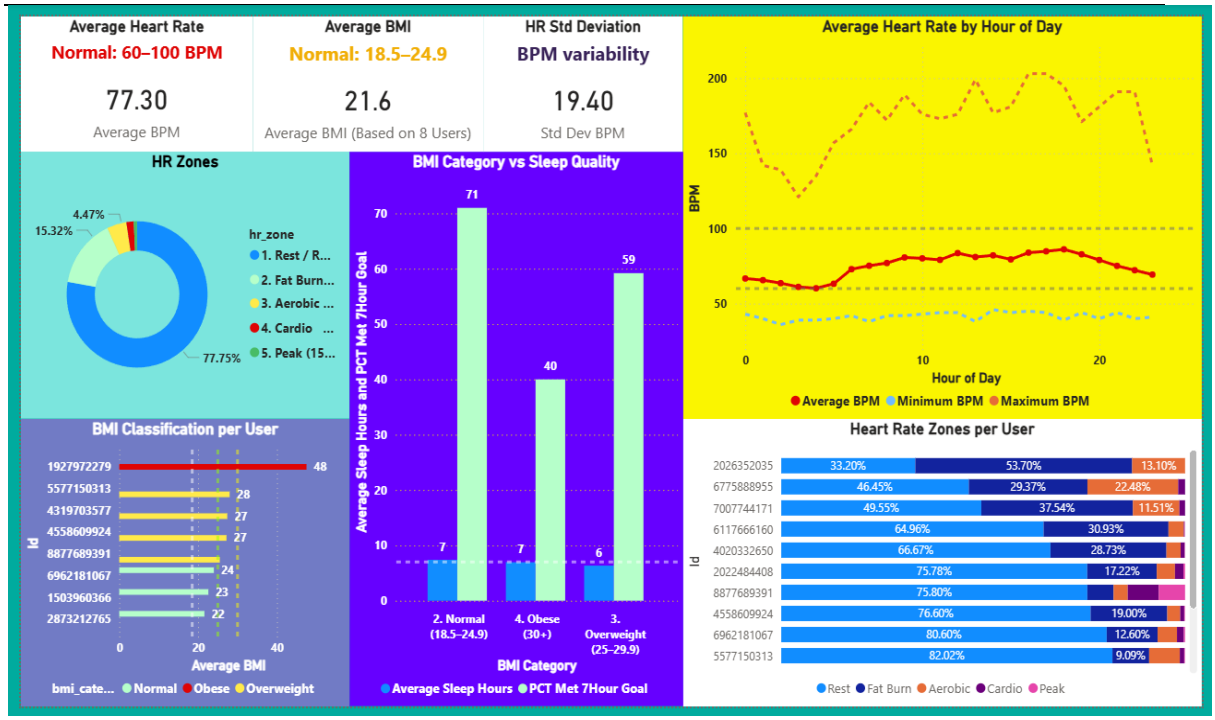


Figure 8: Heart Rate Dashboard: HR by Hour, BMI Distribution, HR Zones, BMI vs. Sleep

7.1 Resting and Daily Heart Rate

Average heart rate across the cohort is 77.3 BPM, within the healthy adult normal range (60-100 BPM). The standard deviation of 19.4 BPM indicates high variability, reflecting the mixed activity composition of the user base. Heart rate follows a predictable circadian pattern with lowest in early morning hours and peaking around 6-7PM in alignment with the evening activity surge identified in Section 3.2 (Figure 3).

7.2 Hear Rate Zone Distribution

The HR zone pie chart shows a heavily Rest/Recovery dominant profile: 77.75% of all tracked heart rate time in Zone 1 (Rest/Resting), with 15.32% in Zone 2(Fat Burn), 4.47% in Zone 3(Aerobic), and minimal time in Cardio or Peak zones. This reinforces the sedentary finding that the majority of tracked time involves minimal cardiovascular demand.

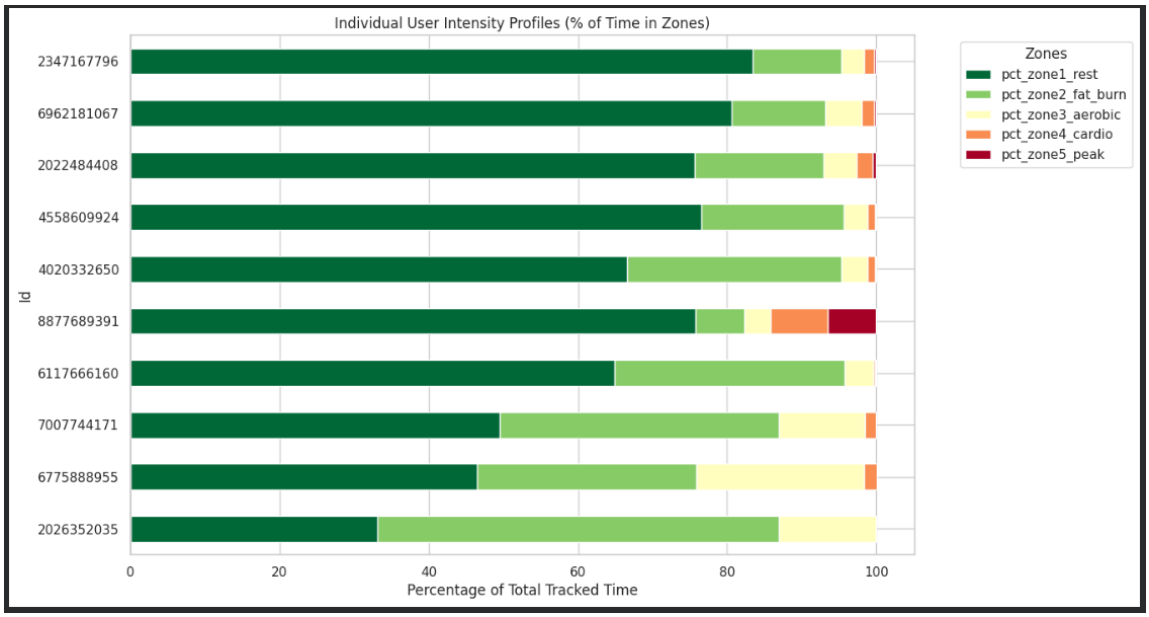


Figure 9: Individual User Intensity Profiles: % of Time in Each HR Zone

Individual user profiles reveal significant variation. User 8877689391 is a notable outlier, spending substantial time in Cardio and Peak zones who is likely and endurance athlete or high intensity exerciser. In contrast, users 2347167796 and 6962181067 spend over 80% of their time in Zone 1, indicating predominantly sedentary activity profiles.

This per-user zone profiling enables targeted content delivery: endurance training tips for high zone users vs. movement prompts and sedentary break reminders for low-zone users.

7.3 BMI Profile

Average BMI is 21.6, within the healthy normal range (18.5-24.9). The BMI classification chart reveals one significant outlier (user 1927972279 with BMI ~48, classified as Obese), while most other users cluster in the Normal and Overweight categories. BMI vs Sleep data shows Normal BMI users averaging 7 hours of sleep vs Overweight users at 7 hours and Obese at 6 hours. These marginal differences may be confounded by other variables.

8. Feature Engagement and Platform Usage

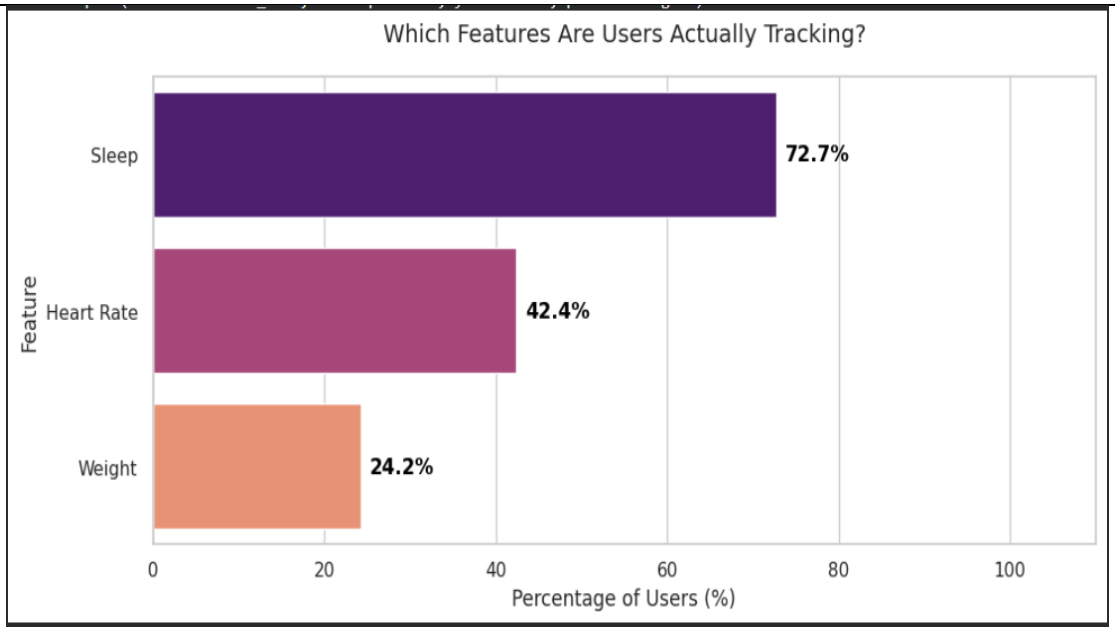


Figure 10: Feature Adoption: % of Users Tracking Sleep, Heart Rate & Weight

Feature adoption data reveals a clear engagement hierarchy among the three optional tracking modules:

- Sleep tracking: With 72.7% of users, this is the most widely adopted supplementary feature.
- Heart rate monitoring: Moderate adoption with 42.4% users that are likely driven by device capability constraints.
- Weight logging: The least adopted feature with 24.2% of users, suggesting a friction in manual data entry.

The gap between sleep (72.7%) and weight (24.2%) adoption is striking and likely reflects the passive vs active nature of data collection. Sleep and heart rate are tracked automatically, but weight requires deliberate user action. This has direct product implications for friction reduction and default on feature design.

Tracker engagement by usage tier is heavily skewed toward high engagement: 27 of 33 users (82%) are in the ‘High’ tier (28-31 days tracked), with only 3 Low (14-20 days), 2 Moderate, and 1 Very Low user. This cohort likely represents self-selected, highly motivated users which reveals a selection bias that should be noted when generalising findings.

9. Strategic Recommendations

1. **Evening Push Notification**
   Peak activity occurs at 6-7PM, therefore scheduling motivational prompts and challenges during this window may maximise engagement and step count conversions.

2. **Tiered Step Goals**
   Replace the one size fits all 10K goal with segment specific targets (5K for sedentary, 7.5K for Somewhat Active). Small wins drive habit formation better than distant targets.

3. **Sedentary Break Reminders**
   With average sedentary time at 15.9 hours/day, hourly movement prompts during 9AM-5PM could materially reduce sedentary time and improve both calorie burn and sleep quality.

4. **Sleep Coaching Programme**
   Only 28.3% meet the 8 hour sleep goal. A guided sleep hygiene programme with bedtime reminders, wind down routines, and sedentary reduction coaching would target the sleep sedentary correlation found in section 4.4 (Figure 7).

5. **Weight Tracking Friction Reduction**
   At 24.2% adoption, weight logging lags behind other features. Implement smart scale integrations, simplified entry UX, auto prompted weekly weigh-ins to increase passive data capture.

6. **Heart Rate Zone Coaching**
   77.75% of heart rate time is in Zone 1. Introduce personalised cardio targets and workout recommendations for users with low zone diversity to drive health outcomes and app engagement.

7. **Thursday Energy Programme**
   Thursday shows the week's lowest step and calorie metrics. A mid-week challenge or 'Boost Thursday' campaign could address this consistent dip.

8. **Outlier User Programmes**
   User 8877689391 (elite HR zones) and User 1927972279 (BMI ~ 480 represent extreme profiles. Develop premium performance content for elite users and clinical grade intervention pathways for high-risk users.

9. Limitations and Data Quality

- A sample size of 33 users limits statistical significance, and therefore findings should be validated with a larger cohort before major product decisions.
- User cohort represents highly engaged, self selected Fitbit owners (82% in high usage tier), introducing selection bias. Results may not generalise to casual users or the general population.
- The sleep activity relationship (Section 4.3) shows counterintuitive results that suggest confounding variables. Causal claims should not be made without controlled analysis.

10. Conclusion

The Fitbit user behaviour analysis reveals a platform with strong engagement fundamentals with 82% of users track consistently across the full month but significant health behaviour gaps relative to established benchmarks. Steps, sleep, and sedentary time are the three areas requiring the most urgent product intervention.

The data supports a shift from passive tracking to active, personalized coaching. Targeted notifications timed to peak activity windows, tiered goal setting that meets users where they are, and evidence-based sleep improvement programmes represent the highest leverage opportunities to drive measurable health outcomes while deepening product engagement.

As the dataset grows, the correlations identified here are particularly between sedentary behaviour and sleep quality, and between activity segment and calorie distribution. It should be validated at scale to inform machine learning models and personalised recommendation engines.